

Theory of Spectral Relativity — Version 3

Annex: Rigidity Observation of the Central Object

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Abstract

This annex documents a central observation that emerged from the RST research programme: the mathematical object $\Phi(x) = \tanh(\alpha \cdot x)$ — equivalently $\rho(x) = \text{sech}^2(\alpha \cdot x)$ — was independently recovered across 14 distinct mathematical and physical frameworks. None of these frameworks assumed it; each reconstructed it. This convergence constitutes a form of mathematical rigidity whose significance warrants explicit statement.

The Central Object

The object in question admits three equivalent representations:

$$\Phi(x) = \tanh(\alpha \cdot x) \text{ [kink profile]}$$

$$\rho(x) = \alpha \cdot \text{sech}^2(\alpha \cdot x) \text{ [spectral density]}$$

$$\psi_0(x) = \text{sech}^2(\alpha \cdot x) \text{ [Goldstone mode]}$$

These three forms are related by $\rho = \Phi'$ and $\psi_0 = \rho/\alpha$. They describe the same object in different languages.

The 14 Frameworks of Reconstruction

In each of the following frameworks, the central object was recovered without being assumed — either as the unique solution satisfying the framework's conditions, or as the emergent canonical structure.

1. KPP / Reaction-diffusion

Test applied: Fixed point of the coupled KPP flow (Eq. 3-4, V3)

Object recovered: $\rho^* = (3\lambda/2\gamma) \cdot \text{sech}^2(\alpha \cdot x)$ exactly

2. Calculus of variations

Test applied: Unique minimiser of $A[\Phi]$ with symmetric double-well potential

Object recovered: $\Phi = \tanh$, unique monotone finite-action solution (Theorem 1.1)

3. Integrable ODE / Bogomolny

Test applied: Autonomous first-order equation $\Phi' = f(\Phi)$

Object recovered: $f(\Phi) = \alpha(1-\Phi^2)$, solution $\tanh$ by quadrature

4. Spectral theory / Poschl-Teller

Test applied: Hessian of $A[\Phi]$ around the kink (Theorem 2.1)

Object recovered: $L = -d^2/dx^2 + 4\alpha^2 - 6\alpha^2 \operatorname{sech}^2(\alpha \cdot x)$

5. Sturm-Liouville theory

Test applied: Positive nodeless zero mode of L

Object recovered: $\psi_0 = \operatorname{sech}^2$: unique nodeless ground state

6. Darboux factorisation / SUSY-QM

Test applied: Reconstruction $L \rightarrow \Phi \rightarrow V$ from the zero mode

Object recovered: $\psi_0 = \operatorname{sech}^2 \Rightarrow \Phi = \tanh \Rightarrow$ quartic V

7. Morse theory

Test applied: Morse index of L (number of negative eigenvalues)

Object recovered: Index = 0: stable kink, spectral gap = $4\alpha^2$

8. Differential geometry

Test applied: Heteroclinic connection in the phase plane (Φ, ρ)

Object recovered: Unique orbit from $(-1,0)$ to $(+1,0)$ on the manifold $H=0$

9. Pohozaev identities

Test applied: Universal integrals over the heteroclinic trajectory

Object recovered: $I_1/I_2 = 5/4$ (pure invariant $J_2 = 6/5$, Section 2.2)

10. Dynamical systems

Test applied: Gradient flow fixed point attractor

Object recovered: Unique monotone heteroclinic connection (Cauchy-Lipschitz)

11. Riemannian spectral geometry

Test applied: Metric $ds^2 = d\omega^2/\rho$ on the spectral manifold (Section 8)

Object recovered: Flat geodesic, isometric to a geodesic of H^2

12. Yang-Mills / QCD

Test applied: IR propagator, string tension, critical temperature

Object recovered: $\sigma = 0.194 \text{ GeV}^2$ (<0.5% lattice), $T_c = 156.7 \text{ MeV}$ (1.1%)

13. Navier-Stokes / Turbulence

Test applied: Spectral flux $\Pi(k)$, JHTDB validation at $Re_\lambda = 433$

Object recovered: $\alpha = 0.93-0.96$, RMS = 0.016, temporal stability <0.1%

14. Pure invariants

Test applied: Constants derived from the kink profile alone

Object recovered: $J_1=6$, $J_2=6/5$, $J_3=1/\sqrt{3}$ — universal, no free parameters

Observation

Across 14 frameworks tested, 14 recovered the same object. No failure was recorded. One additional domain (marginal glasses, Section 15 of V3) is identified as a candidate but has not yet been tested.

This convergence is not presented as a theorem in the classical sense. It constitutes a rigidity observation: the central object of RST resists every change of mathematical representation. Each framework that approaches it recovers it, regardless of the entry point.

Frameworks 1 through 11 are mathematical in nature. Frameworks 12 and 13 constitute experimental validations on real data (QCD lattice and JHTDB database). Framework 14 produces invariants independently verifiable from any physical domain.

Status Note

Consistent with the RST anti-circularity protocol, this annex distinguishes what is established from what remains open. Results from frameworks 1-11 and 14 are verifiable via the V3 verification scripts or by direct computation. Results from frameworks 12-13 are reproducible from public JHTDB data and the QCD lattice references cited in the main text.

A formal characterisation theorem — demonstrating that these 14 representations are exhaustive and that each necessarily reconstructs the same object — remains an open problem. The observation documented here provides its empirical motivation.